

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied business analytics

Self Study

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Date of Assignment:24-07-2026	Date/Time of Submission:24-07-2026

Q.1 Write a Python program to:

- Find the Top 10 Genres.
- Find the Bottom 10 Genres.
- Identify the most popular Genre.
- Identify the least popular Genre.
- Display the result using a bar chart.

Python

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("NetFlix.csv")

# Clean comma-separated genres
genres = df['genres'].dropna().str.split(', ').explode()
genre_counts = genres.value_counts()

top_10_genres = genre_counts.head(10)
bottom_10_genres = genre_counts.tail(10)
most_popular = genre_counts.idxmax()
least_popular = genre_counts.idxmin()

print("--- Top 10 Genres ---")
print(top_10_genres)

print("\n--- Bottom 10 Genres ---")
```

```

print(bottom_10_genres)

print(f"\nMost Popular Genre: {most_popular} ({genre_counts.max()}\ntitles)")
print(f"Least Popular Genre: {least_popular} ({genre_counts.min()}\ntitles)")

# Visualization
plt.figure(figsize=(10, 5))
top_10_genres.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Top 10 Netflix Genres')
plt.xlabel('Genre')
plt.ylabel('Number of Titles')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

```

Output:

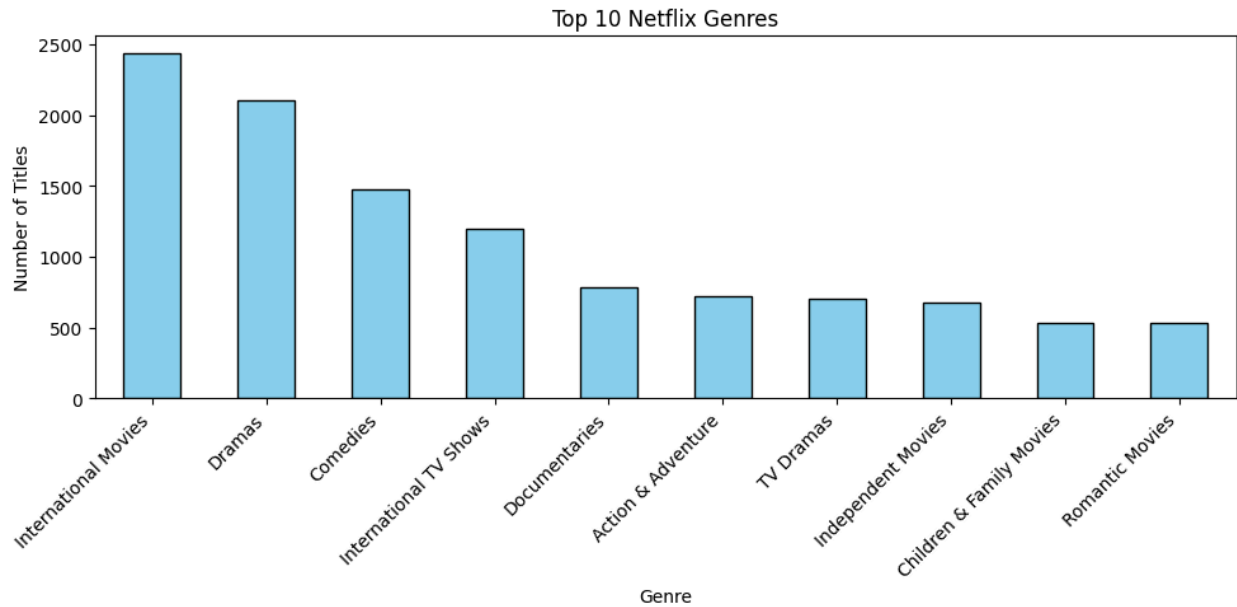
```

--- Top 10 Genres ---
genres
International Movies      2437
Dramas                   2106
Comedies                  1471
International TV Shows    1199
Documentaries             786
Action & Adventure        721
TV Dramas                 704
Independent Movies        673
Children & Family Movies  532
Romantic Movies           531
Name: count, dtype: int64

--- Bottom 10 Genres ---
genres
TV Horror                 69
Teen TV Shows             60
Cult Movies               59
Faith & Spirituality      57
Anime Features            57
Movies                   56
Stand-Up Comedy & Talk Shows 52
TV Thrillers              50
Classic & Cult TV         27
...
Name: count, dtype: int64

Most Popular Genre: International Movies (2437 titles)
Least Popular Genre: TV Shows (12 titles)

```



Q.2 Write a Python program to find:

- The oldest Movie.
- The newest Movie.
- The oldest TV Show.
- The newest TV Show.

Display all details of these titles.

Python

```
import pandas as pd
```

```
df = pd.read_csv("NetFlix.csv")
```

```
# Filter by type
```

```
movies = df[df['type'].str.strip() == 'Movie']
```

```
tv_shows = df[df['type'].str.strip() == 'TV Show']
```

```
oldest_movie = movies[movies['release_year'] ==  
movies['release_year'].min()]
```

```
newest_movie = movies[movies['release_year'] ==  
movies['release_year'].max()]
```

```

oldest_tv = tv_shows[tv_shows['release_year'] ==
tv_shows['release_year'].min()]
newest_tv = tv_shows[tv_shows['release_year'] ==
tv_shows['release_year'].max()]

print("=== OLDEST MOVIE ===")
print(oldest_movie[['title', 'type', 'release_year',
'genres']].to_string(index=False))

print("\n=== NEWEST MOVIE ===")
print(newest_movie[['title', 'type', 'release_year',
'genres']].head(1).to_string(index=False))

print("\n=== OLDEST TV SHOW ===")
print(oldest_tv[['title', 'type', 'release_year',
'genres']].to_string(index=False))

print("\n=== NEWEST TV SHOW ===")
print(newest_tv[['title', 'type', 'release_year',
'genres']].head(1).to_string(index=False))

```

Output:

```

=== OLDEST MOVIE ===
      title  type  release_year      genres
Prelude to War Movie      1942 Classic Movies, Documentaries
The Battle of Midway Movie      1942 Classic Movies, Documentaries

=== NEWEST MOVIE ===
      title  type  release_year      genres
Charming Movie      2021 Children & Family Movies, Comedies

=== OLDEST TV SHOW ===
      title  type  release_year      genres
Pioneers: First Women Filmmakers* TV Show      1925 TV Shows

=== NEWEST TV SHOW ===
      title  type  release_year      genres
Carmen Sandiego TV Show      2021 Kids' TV, TV Thrillers

```

Q.3 Write a Python program to:

- Find the country with the highest Netflix content.

- Display the Top 10 Countries.
- Display the Bottom 10 Countries.
- Display the result using a horizontal bar chart.

Python

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("NetFlix.csv")

countries = df['country'].dropna().str.split(', ').explode()
country_counts = countries.value_counts()

top_country = country_counts.idxmax()
top_10_countries = country_counts.head(10)
bottom_10_countries = country_counts.tail(10)

print(f"Country with Highest Content: {top_country}
({country_counts.max()} titles)\n")

print("--- Top 10 Countries ---")
print(top_10_countries)

print("\n--- Bottom 10 Countries ---")
print(bottom_10_countries)

# Horizontal Bar Chart
plt.figure(figsize=(10, 6))
top_10_countries.sort_values(ascending=True).plot(kind='barh',
color='crimson')
plt.title('Top 10 Countries by Netflix Content')
plt.xlabel('Number of Titles')
plt.ylabel('Country')
plt.tight_layout()
plt.show()
```

Output:

Country with Highest Content: United States (3296 titles)

--- Top 10 Countries ---

country

United States	3296
India	990
United Kingdom	722
Canada	412
France	349
Japan	287
Spain	215
South Korea	212
Germany	199
Mexico	154

Name: count, dtype: int64

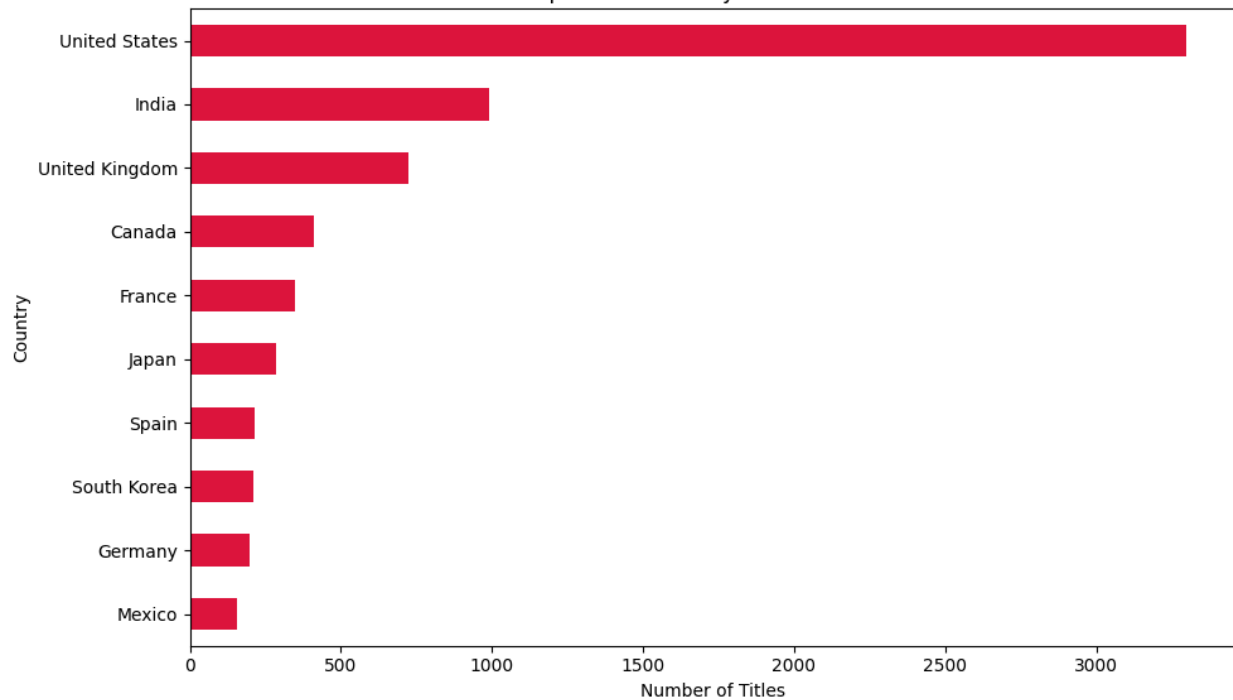
--- Bottom 10 Countries ---

country

Somalia	1
Sudan	1
Panama	1
Armenia	1
Mongolia	1
United Kingdom,	1
Uganda	1
East Germany	1
Montenegro	1
Syria	1

Name: count, dtype: int64

Top 10 Countries by Netflix Content



Q.4 Write a Python program to:

- Find the director with the highest number of Movies.
- Find the director with the highest number of TV Shows.
- Display the Top 10 Directors.
- Plot a bar chart of the Top 10 Directors.

Python

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("NetFlix.csv")

# Directors by Movies
movies_df = df[(df['type'].str.strip() == 'Movie') &
               (df['director'].notna())]
movie_directors = movies_df['director'].str.split(',').explode().value_counts()
top_movie_dir = movie_directors.idxmax()

# Directors by TV Shows
tv_df = df[(df['type'].str.strip() == 'TV Show') &
           (df['director'].notna())]
tv_directors = tv_df['director'].str.split(',').explode().value_counts()
top_tv_dir = tv_directors.idxmax()

# Top 10 Directors Overall
all_directors = df['director'].dropna().str.split(',').explode().value_counts()
top_10_directors = all_directors.head(10)

print(f"Top Movie Director: {top_movie_dir} ({movie_directors.max()} movies)")
print(f"Top TV Show Director: {top_tv_dir} ({tv_directors.max()} TV shows)\n")

print("--- Top 10 Directors Overall ---")
```

```

print(top_10_directors)

# Bar Chart
plt.figure(figsize=(10, 5))
top_10_directors.plot(kind='bar', color='teal', edgecolor='black')
plt.title('Top 10 Directors on Netflix')
plt.xlabel('Director')
plt.ylabel('Number of Titles')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

```

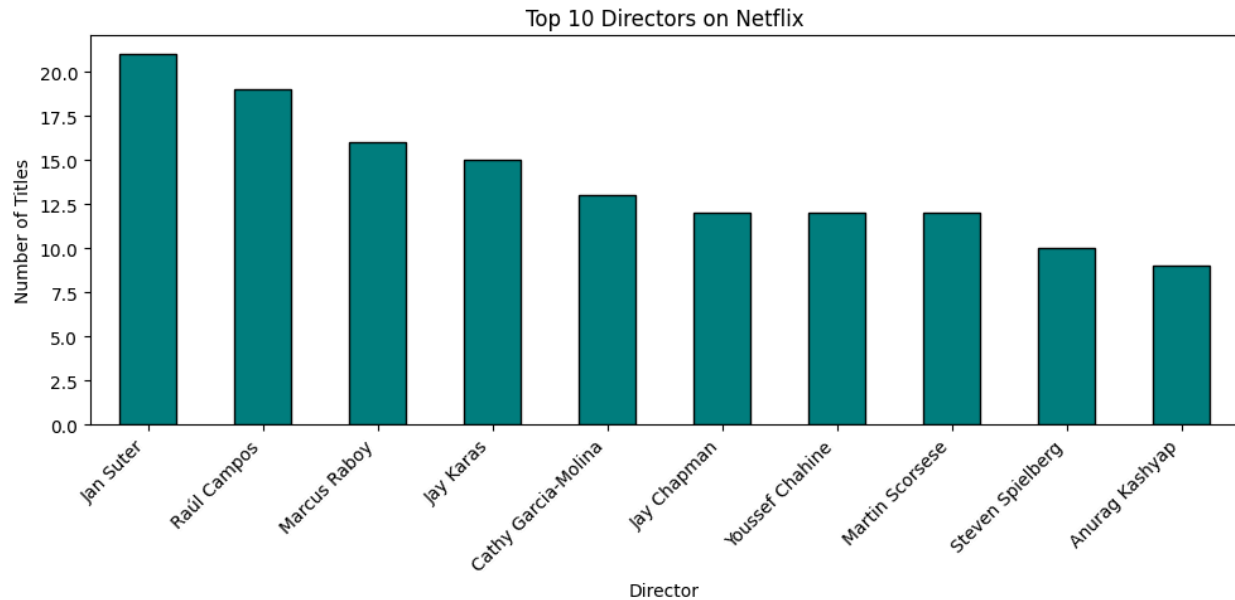
Output:

```

Top Movie Director: Jan Suter (21 movies)
Top TV Show Director: Alastair Fothergill (3 TV shows)

--- Top 10 Directors Overall ---
director
Jan Suter          21
Raúl Campos        19
Marcus Raboy       16
Jay Karas          15
Cathy Garcia-Molina 13
Jay Chapman        12
Youssef Chahine    12
Martin Scorsese    12
Steven Spielberg   10
Anurag Kashyap     9
Name: count, dtype: int64

```



Q.5 Compare Netflix releases:

- Before 2020.
- During 2020.
- After 2020.
- Display the result using a bar chart or line chart.

Python

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("NetFlix.csv")

before_2020 = (df['release_year'] < 2020).sum()
during_2020 = (df['release_year'] == 2020).sum()
after_2020 = (df['release_year'] > 2020).sum()

comparison = pd.Series({
    'Before 2020': before_2020,
    'During 2020': during_2020,
    'After 2020': after_2020
})
```

```

print("--- Netflix Releases Comparison ---")
print(comparison)

# Bar Chart
plt.figure(figsize=(8, 5))
bars = plt.bar(comparison.index, comparison.values,
color=['#4C72B0', '#DD8452', '#55A868'], edgecolor='black')

for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 20,
f'{yval:,.}', ha='center', va='bottom', fontweight='bold')

plt.title('Netflix Releases: Before, During, and After 2020')
plt.xlabel('Time Period')
plt.ylabel('Number of Releases')
plt.ylim(0, max(comparison.values) * 1.15)
plt.tight_layout()
plt.show()

```

Output:

Netflix Releases: Before, During, and After 2020

